TCV-ϕ III: primordial perturbations across the cohesive bounce and CLASS/CAMB bridge diagnostics

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This third paper develops the primordial perturbation sector of the same TCV-ϕ EFT defined in Papers I and II. Our scope is to formulate and benchmark the Mukhanov–Sasaki pipeline across a cohesive-bounce background, compute tabulated primordial spectra, and extract baseline observables such as (n_s, A_s, r, α_s) in explicit benchmark regimes. At this stage we distinguish clearly between EFT-derived ingredients and phenomenological closure ansätze used to complete the background and perturbative calculations. We also provide a reproducible bridge to CLASS/CAMB using exported primordial tables, as a consistency tool rather than a full likelihood-level fit.

I. INTRODUCTION

Paper I defined the canonical TCV-ϕ EFT and established a conservative late-time cosmology baseline. Paper II introduced and validated the numerical pipelines for bounce/reheating/baryogenesis and ULDM background diagnostics. The next logical step is the primordial perturbation sector.

The objective of this paper is therefore to provide a PRD-ready, reproducible treatment of scalar (and optionally tensor) perturbations across the cohesive bounce. The emphasis is methodological and falsifiable: all claims are tied to explicit numerical diagnostics exported by scripts in `papers/paper-03/code/`.

II. CANONICAL EFT BASELINE AND CONVENTIONS

We use exactly the same EFT definitions, notation, and benchmark hierarchy as in Papers I and II. No new fundamental fields are introduced in this paper. The CLASS/CAMB cosmological baseline is kept fixed to the exact canonical tuple of Paper I, without retuning in Paper III.

III. BOUNCE BACKGROUND AND PERTURBATION VARIABLES

We consider a cohesive-bounce background consistent with the Paper II pipeline. Where the background profile is not uniquely fixed by the EFT, we explicitly label the closure as a phenomenological ansatz. For the benchmark shown here, the primordial run uses the same cohesive-mixture closure as in Paper II:

$$w_{\rm eff} = -0.003, \qquad C_0 = -0.0038, \qquad \eta_c = 50, \qquad \texttt{mode} = \texttt{lorentzian}, \tag{1}$$

with $(\eta_i, \eta_f) = (-1000, -1)$ and a 60-mode logarithmic k grid in the interval $10^{-3} \leq k \leq 10^{-1}$. The parameter $d\phi_{\rm rel} = 10^{-5}$ controls the effective cohesive profile used in $z(\eta)$ and does not replace the Mukhanov–Sasaki vacuum initial conditions.

IV. MUKHANOV–SASAKI PIPELINE

Scalar modes are evolved with

$$v_k'' + \left(c_s^2 k^2 - \frac{z''}{z}\right) v_k = 0, \tag{2}$$

where for the present benchmark we set $c_s = 1$, consistently with the Paper I late-time regime where non-minimal corrections from $M_{\rm eff}(\Phi_1)$ are negligible. Quantitatively, this follows the Paper I estimate $\Delta M_{\rm eff}^2/M_{\rm eff}^2 \sim 10^{-16}$ for the canonical benchmark, which makes deviations from the canonical scalar propagation negligible at the level of the present setup. Initial conditions are chosen as adiabatic Bunch–Davies vacuum conditions at η_i, $v_k(\eta_i) = e^{-ik\eta_i}/\sqrt{2k}$ and $v_k'(\eta_i) = -i\sqrt{k/2}\, e^{-ik\eta_i}$.

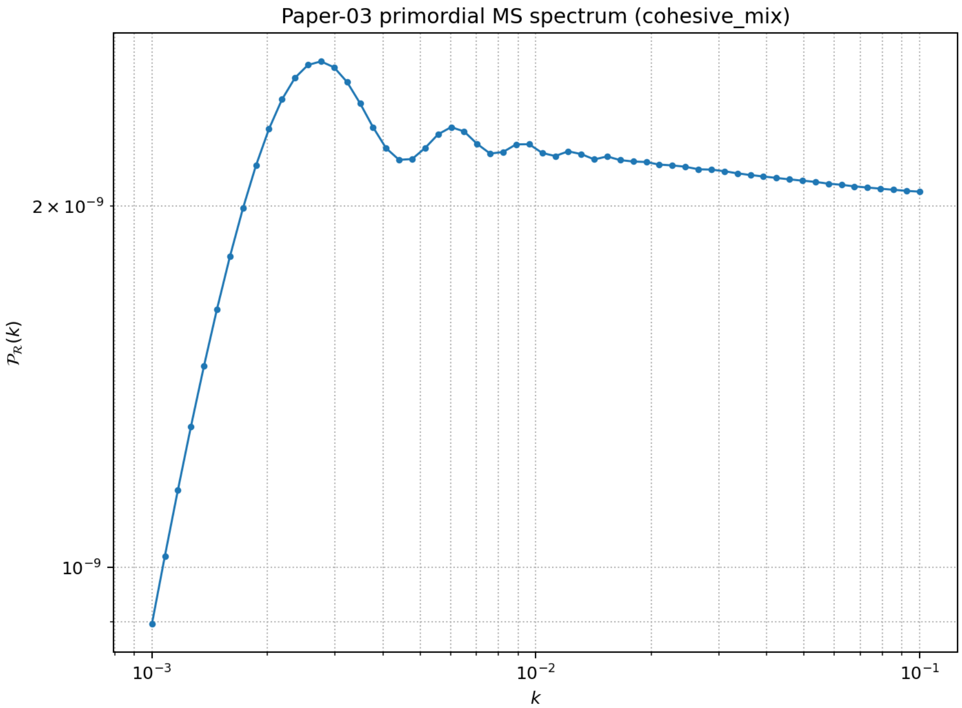

FIG. 1. Primordial scalar spectrum generated by the Paper III MS pipeline (`compute_ms_spectrum.py`) for the cohesive-mixture benchmark.

The pipeline exports tabulated primordial spectra

$$\mathcal{P}_{\mathcal{R}}(k) \tag{3}$$

into `papers/paper-03/data/PR_tcvphi.dat`. For this benchmark, the numerical fit from the MS solver gives

$$n_s^{\mathrm{MS}} \simeq 0.9785, \tag{4}$$

from a broad-band fit over the full tabulated k interval, before the local pivot fit step used by the Boltzmann bridge. The raw MS spectrum is then rescaled by a constant factor to match the target amplitude $A_s = 2.1 \times 10^{-9}$ at $k_{\mathrm{pivot}} = 0.05$; for this run the pre-rescaling amplitude is $A_s^{\mathrm{raw}} \simeq 2.80 \times 10^{-2}$.

V. PRIMORDIAL OBSERVABLES AND BENCHMARK WINDOWS

From the computed spectra we extract baseline observables:

$$(n_s,\, A_s,\, r,\, \alpha_s), \tag{5}$$

within explicitly stated benchmark windows. For the current scalar-only benchmark run, we obtain

$$A_s \simeq 2.1 \times 10^{-9}, \tag{6}$$

$$n_s \simeq 0.9643 \quad \text{(local fit around } k_{\mathrm{pivot}} = 0.05\text{)}, \tag{7}$$

$$\alpha_s \simeq -1.43 \times 10^{-3}, \tag{8}$$

with r not extracted at this stage by the scalar-only script. The difference between n_s^{MS} and the quoted local-fit value reflects two different estimators: a global fit over the full k range versus a local polynomial fit around the pivot, the latter being the quantity propagated to the CLASS external-spectrum bridge. In the current script the local quadratic fit uses the 35 points closest to k_{pivot} (configurable with `-fit-window`).

Observable	Benchmark value
A_s	2.1×10^{-9}
n_s (local fit)	0.9643
α_s (quadratic fit)	-1.43×10^{-3}
r	not extracted in scalar-only run

TABLE I. Paper III baseline primordial observables for the current benchmark.

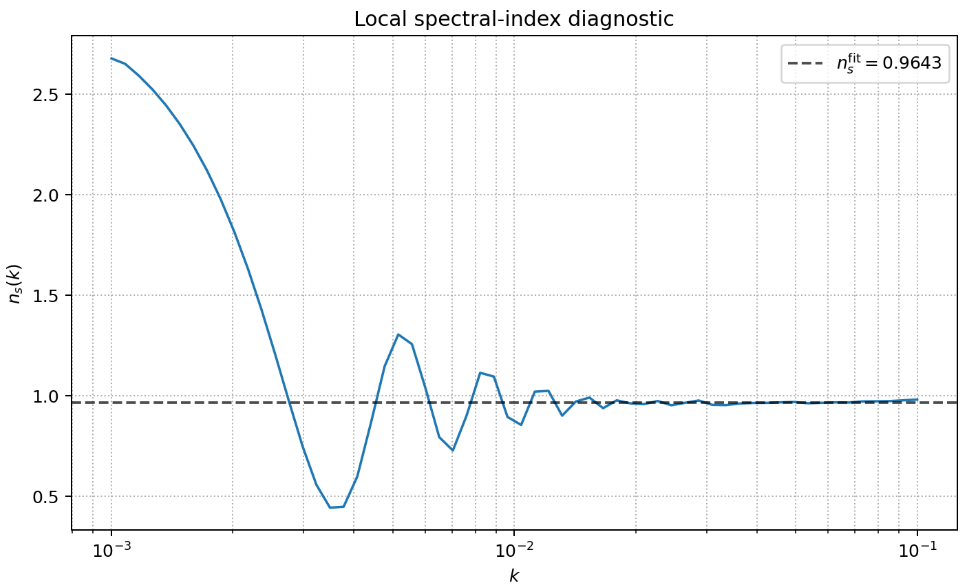

FIG. 2. Local spectral-index diagnostic derived from the tabulated primordial spectrum (`compute_primordial_observables.py`).

VI. CLASS/CAMB BRIDGE STATUS AND CONSISTENCY CHECKS

Tabulated primordial spectra are passed to the shared TCV-ϕ Boltzmann interface. In this paper, these runs are used as reproducibility and consistency checks. A full data-level likelihood analysis is deferred to dedicated follow-up work. In the present benchmark, CLASS runs in both modes (power-law and external table) complete without fallback, and the ratio of reconstructed matter spectra remains close to unity:

$$0.9988 \lesssim \frac{P_{\rm external}(k)}{P_{\rm powerlaw}(k)} \lesssim 1.0014. \tag{9}$$

This confirms interface-level consistency of the exported primordial table for the chosen benchmark. CAMB runs are not included in the present benchmark script and are left to a dedicated follow-up implementation.

VII. STATUS OF ASSUMPTIONS AND OUTLOOK

This paper keeps the same tiered status labelling as Papers I and II:

- EFT-derived equations and definitions;
- phenomenological closure ansätze used for numerical completion;
- deferred sectors for later papers (e.g. full strong-gravity BH analysis in Paper IV).

To keep parameter consistency explicit across papers, each run writes a JSON summary that stores both (i) canonical Paper I CLASS/CAMB inputs and (ii) the Paper III ansatz controls used by the MS solver.

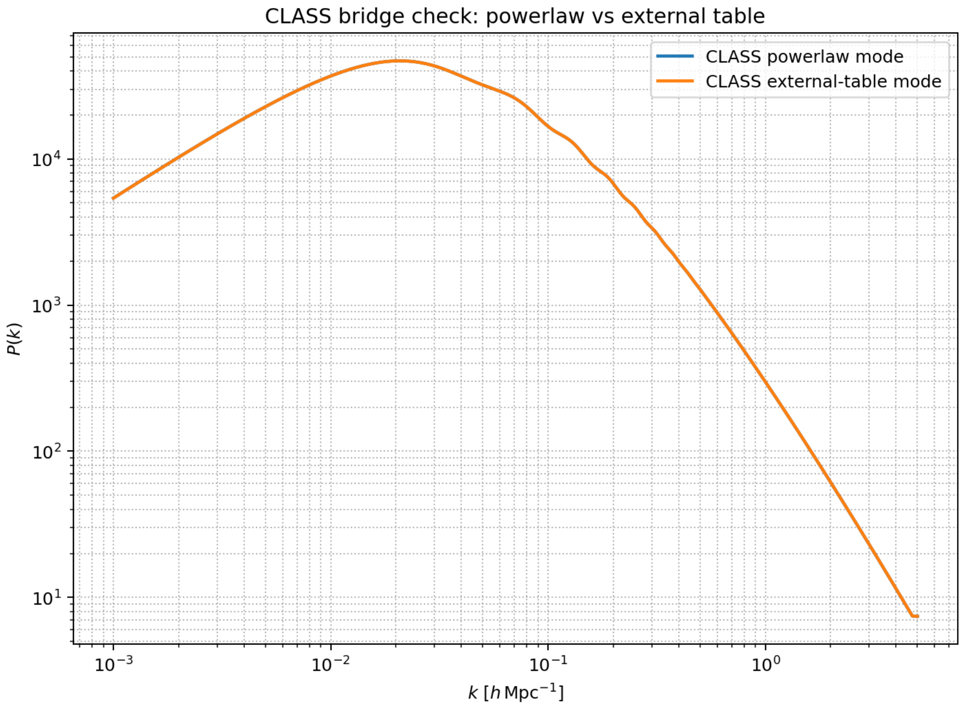

FIG. 3. CLASS bridge consistency check: power-law mode versus external primordial-table mode.

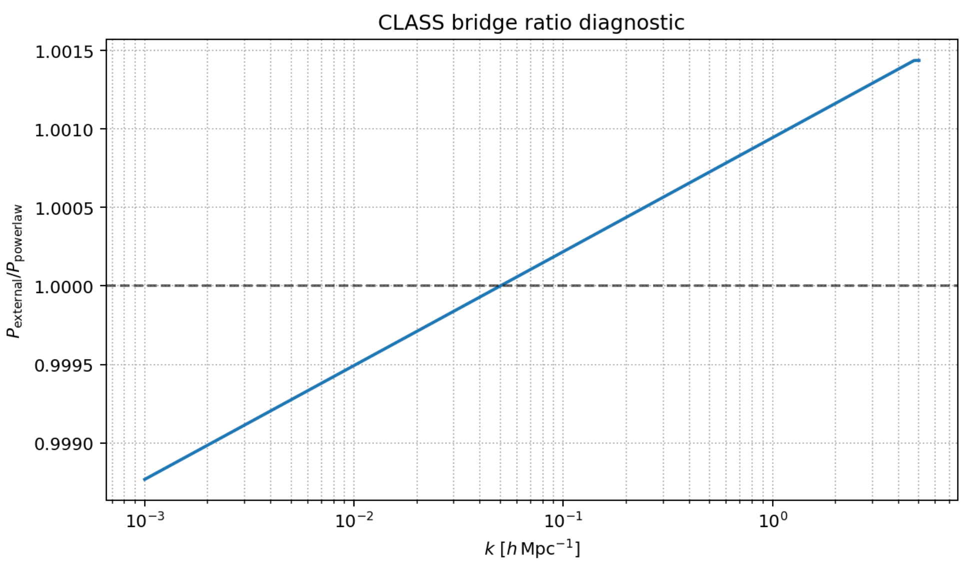

FIG. 4. Ratio diagnostic for the CLASS bridge test. The near-unity ratio is expected for the matched benchmark setup and validates table-mode reproducibility.

VIII. CONCLUSION

Paper III delivers the primordial perturbation layer of the TCV-ϕ programme in a reproducible PRD-ready format. The results are presented conservatively, with explicit diagnostics and without claims beyond the demonstrated benchmark scope.

[1] C. Lecroq, "TCV-ϕ I: an effective cohesive–vacuum field theory and its ΛCDM–compatible background cosmology," preprint (2026), hal-05547048.

[2] C. Lecroq, "TCV-ϕ II: cohesive bounce, geometric reheating, and baseline baryogenesis/ULDM pipelines," preprint (2026), hal-05547079.